Dr. Sarah Wassermann

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AI/ML & Cybersecurity | International experience in France, Austria, and the United States

Experience

Research Scientist @ Proofpoint (formerly Hornetsecurity/Vade) **Paris (FR) | August 2022 – April 2026**

Advanced **email security** through **NLP**, Large Language Models (**LLMs**), and Machine Learning **(ML)**.

- Deployed a **RAG-based LLM cybersecurity training agent**, handling up to 1M queries/month under 10 s latency SLOs, by **leading end-to-end design** (retrieval pipeline based on vector search, integration of hosted models, fault-tolerant architecture, evaluation using LLM-as-a-judge). Acted as **product owner** to align research roadmap with security and company-wide training KPIs of our clients.
- Architected **an enterprise-grade RAG application** for automated legal document analysis, engineering retrieval pipelines to surface cross-corpus inconsistencies and resolve complex, multi-hop queries. **Steered the product roadmap**, translating client operational requirements into AI research initiatives.
- **Improved email threat detection** by applying NLP, LLMs, and ML to large-scale email streams, improving malicious email detection accuracy (**90%+ detection rate**) compared to prior rules-based systems.
- **Tech Lead** for a critical **data clustering project** for dynamic email datasets to boost detection accuracy and eliminate the need for manual review.
- Pioneered an **Explainable AI (XAI)** initiative, creating a tool to interpret AI model decisions for end-users.
- **Tech stack:** Python, scikit-learn, NLTK, ollama, LlamaIndex, Pandas, NumPy, FastAPI, MongoDB, Chroma, Git, Gitlab CI.

Research Engineer @ Huawei **Paris (FR) | Sept 2021 – March 2022**

Contributed to projects related to **hardware-failure prediction (predictive maintenance)** for hyperscale data centers and to estimation of **Customer Experience Index (CEI).**

- Won the **Rising Star Award 2021** to recognize my exceptional contributions and rapid impact within the research group.
- Engineered the entire Python **ML pipeline**, implementing **advanced feature selection** and **XAI** for model **interpretability. Reduced feature set by 70%** while keeping **accuracy above 80%**.
- Enhanced **CEI estimation** by designing and implementing complex **KPI-to-KQI mappings**, aligning network performance metrics with the business goal of end-user satisfaction.
- **Tech stack:** Python, scikit-learn, Pandas, NumPy, Git, Gitlab CI.

Education & Research Internships

PhD. in Computer Science @ TU Wien & AIT **Vienna (AT) | Oct 2017 – May 2022**

Developed ML algorithms for **advanced network traffic analysis** on encrypted data streams, contributing to both Quality of Experience (QoE) debugging and network security insights.

- Published **30+ papers** (https://scholar.google.com/citations?user=tCCFBrsAAAAJ) in top conferences (SIGCOMM, NeurIPS, IMC, IEEE TNSM) with **500+ citations** to date and 1 **Best Paper Award**.
- Served as a **reviewer** and **TPC member** for major networking conferences and journals (PAM, IEEE/ACM ToN, IEEE TNSM…).
- **Reduced manual labeling effort** by up to **20%** while maintaining target inference accuracy by engineering a **reinforcement-learning-based active learning algorithm**.
- Implemented a lightweight **ML monitoring system** to infer **video and Web QoE** from encrypted traffic in real-time (less than **1 ms**), achieving **94%+ inference accuracy** and reducing the number of features by a **factor 10** compared to SOTA.

Visiting Researcher @ Northwestern University **Chicago (IL) | July – Sep 2016**

- Analyzed **anycast-based Content Delivery Networks** (CDN) in cellular networks.
- Evaluated **real-world DNS anycast service performance** and diagnosed **causes of poor user experience**.

Research Intern @ FTW **Vienna (AT) | July – Sep 2015**

- Developed DisNETPerf, a system leveraging the RIPE Atlas measurement platform, to **perform active network measurements** (traceroute) for monitoring Internet path performance for ISPs, particularly for CDNs.

Summer Intern @ LIP6 (Sorbonne Université) **Paris (FR) | July 2014**

- Contributed **Paris Ping** to the **libparistraceroute** open-source library to avoid flow-based load balancing, while handling IP (IPv4, IPv6), and TCP, UDP, ICMP probes.

MSc. & BSc. in Computer Science @ ULiège **Liège (BE) | Sep 2012 – Sep 2017**

Ranked in **the top 3** students every year. Worked as an **undergraduate teaching assistant** for programming courses.

Skills

Machine learning and statistics: supervised/unsupervised learning, active learning, XAI, data modeling
Databases: MongoDB, Microsoft SQL Server
Programming languages: Python (primary), C, C++, MATLAB, Java